

The formation of a VOLCANIC ISLAND CHAIN

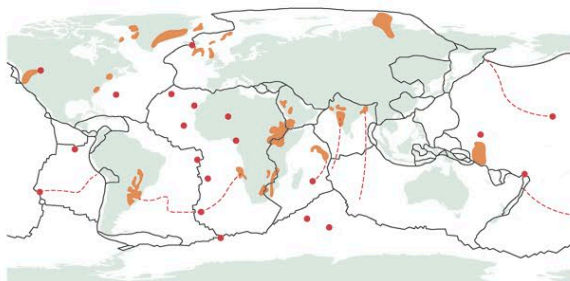
Like beads in a necklace, volcanic islands often form in arcs, or chains, where two oceanic tectonic plates converge or where an oceanic plate meets a continental plate. The volcanic island chains that encircle the Pacific Ocean, which are known as the Ring of Fire, are formed by both processes, as well as, in the case of the Hawaiian Islands, hotspot volcanism.

Hot marks the spot

Hotspots are areas in Earth's mantle that allow magma to rise through the lithosphere until it erupts, often underwater on the seabed. The hotspot that formed the Hawaiian island chain is located in the middle of the Pacific Plate. While the hotspot is fixed, the plate moves continuously. As the plate migrated over the hotspot, the latter created the shield volcanoes responsible for all 132 islands, atolls, reefs, banks, and seamounts of the Hawaiian archipelago. Currently, the hotspot lies beneath the island of Hawaii, where three volcanoes continue to erupt. It also generates activity at the youngest member of the chain, the submarine volcano Loihi, about 19 miles (30 km) south of Hawaii. Loihi last erupted in 1996, following a series of minor earthquakes.

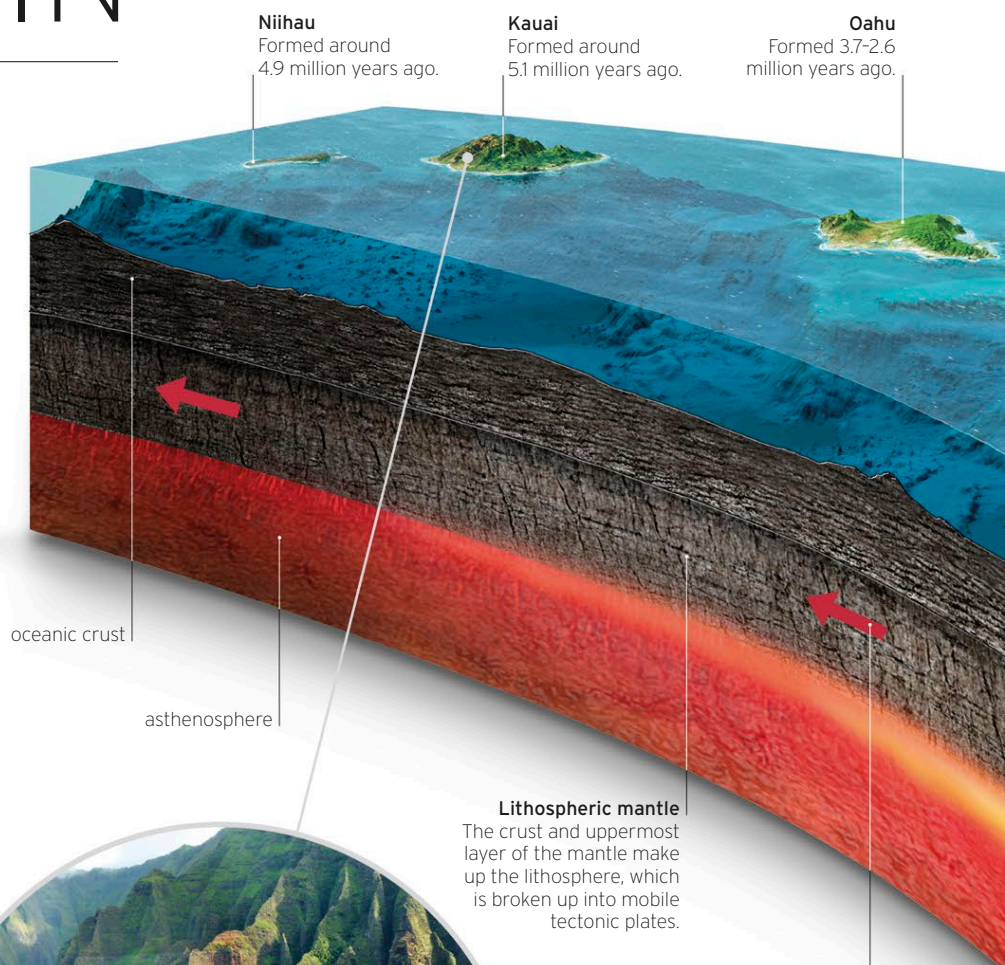
HOTSPOTS AND FLOOD BASALTS

Hotspots are thought to be stationary. As tectonic plates move over them, chains of progressively older volcanoes form in what is known as a hotspot track. The track follows the direction of plate movement, with the youngest, most active volcanoes and seamounts found nearest the hotspot and the less active and extinct ones farthest away. Hotspot tracks are often marked by large areas of basaltic lava, which flooded the area during peak eruption periods.



KEY ● Hotspot
— Hotspot track
■ Flood basalt
— Plate boundary

Kilauea has released enough **lava** to **encircle** Earth three times



Lithospheric mantle

The crust and uppermost layer of the mantle make up the lithosphere, which is broken up into mobile tectonic plates.

Movement of plate

The Pacific Plate moves over the asthenosphere from southeast to northwest at a rate of 2-4 in (5-10 cm) a year.

SCULPTED CLIFFS

The fluted Na Pali cliffs dominate the northwest coast of the island of Kauai, towering up to 4,000 ft (1,200 m) above the Pacific Ocean. The flanks of shield volcanoes often collapse, resulting in jagged, water-permeable surfaces that are further sculpted by rain and wave action.

▲ THE HAWAIIAN ISLANDS

This cross-section shows how conditions in Earth's interior have given rise to the Hawaiian Islands. The Pacific Plate moves over a mantle hotspot, setting off volcanic activity at the surface. With increasing distance from the hotspot, the islands cool and activity diminishes.